

Metasploitable 3 Deployment and Usage Guide

Purpose

Metasploitable 3 is an intentionally vulnerable virtual machine used to simulate real-world exploitation scenarios within the Virtual SOC environment. It provides a high-signal attack surface that generates: - Suricata NIDS alerts - Wazuh SIEM and EDR alerts (optional agent) - Network scans, brute force attempts, and exploitation logs

It functions as the primary blue-team training target for demonstrating detection, logging, and correlation workflows.

1. Image Acquisition

Download the OVA package:

```
https://sourceforge.net/projects/metasploitable3-ub1404upgraded/
```

Extract contents:

```
tar -xvf Metasploitable3.ova
```

This produces one or more VMDK disk files.

2. Convert VMDK to QCOW2 for OpenStack

Use `qemu-img` to convert disk format:

```
qemu-img convert -f vmdk -O qcow2 Metasploitable3-disk001.vmdk  
metasploitable3.qcow2
```

The resulting image is compatible with OpenStack Glance.

3. Upload Image to OpenStack

```
openstack --insecure image create "metasploitable3"
--file metasploitable3.qcow2
--disk-format qcow2
--container-format bare
--private
--property hw_vif_model=e1000 -v
```

Note: The network interface model must be `e1000` for compatibility.

4. Create Instance in VSOC Environment

Create an instance from the uploaded image: - Attach it to **internal-net (105.45.5.0/24)** - Assign DHCP or static IP via pfSense DHCP - Size recommendation: m1.small or m1.medium

Confirm it receives an IP such as:

```
105.45.5.x
```

5. Default Credentials

Metasploitable 3 login:

```
username: vagrant
password: vagrant
```

SSH typically enabled by default.

6. Services Provided by Metasploitable 3

Metasploitable 3 includes a wide variety of vulnerable services:

Network Services

- SSH
- FTP
- SMTP
- DNS

- HTTP / HTTPS
- SMB
- RPC

Web Applications

- Vulnerable PHP applications
- Insecure upload forms
- Broken authentication flows
- Outdated frameworks with RCE flaws

Databases

- MySQL and PostgreSQL with default/weak credentials

Exposed Dev/CI Tools

- Jenkins
- Redis
- Elasticsearch

Auth Weaknesses

- Hardcoded passwords
- Anonymous logins
- Misconfigured sudo rules

Kernel and Package Issues

- Outdated Ubuntu 14.04 kernel
- Numerous known CVEs

7. Integration with SOC Tooling

Suricata NIDS

Metasploitable 3 generates: - Port scan detections - Malware signature hits - Exploit traffic alerts - Brute-force login alerts

Wazuh SIEM (optional agent)

If Wazuh agent is installed (manual): - FIM alerts on system config changes - Authentication failures - Process creation events - Vulnerability scan results

Kali Attacker (offensive workflow)

Kali → Metasploitable 3 produces: - High volume Suricata alerts - SIEM correlation with host logs (if agent exists)

8. Usage Scenarios in VSOC

Scenario 1 — Reconnaissance

```
nmap -sV <meta-ip>
```

Generates SCAN events in Suricata.

Scenario 2 — Weak Credential Exploits

FTP anonymous login or SSH brute-force will trigger: - Suricata: brute-force attempts - Wazuh (if agent installed): failed login logs

Scenario 3 — Web Exploits

Use Metasploit or manual payloads. Produces: - HTTP signature alerts - Possible RCE logs

Scenario 4 — Privilege Escalation

Running known local exploits triggers Wazuh file or process alerts.

Scenario 5 — Malware Deployment

Dropping test malware files (EICAR) triggers SIEM detections (if FIM rules enabled).

9. Safety Considerations

- Never expose Metasploitable 3 to public networks.
 - Keep it isolated in **internal-net**.
 - Disable floating IPs.
 - Use pfSense routing to control attacker access.
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10. Summary of Responsibilities

Metasploitable 3 serves as: - A high-noise, vulnerable victim for attacks - A consistent generator of NIDS alerts - A rich log source for SIEM correlation - A safe practice environment for offensive + defensive exercises

It is the backbone of hands-on exploitation and detection demonstrations within the Virtual SOC.